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DEPARTMENT OF COMPUTER ENGINEERING
CMPE-211 Object Oriented Programming



OOPS LAB REPORT

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Subject:	Object Oriented Programming
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Question no 1:

Create a class Student with private data members name and marks, and a static data member totalStudents to track the total number of students created. Implement a constructor that initializes name and marks and automatically increments totalStudents, and a destructor that prints a message like "Student removed from system". Include a public member function display () to show the student's name and marks. In the main () function, create three to four student objects dynamically using new, display their information, and then delete the dynamic objects, printing the value of totalStudents after each deletion. This will help students understand how constructors, destructors, static members, and dynamic memory work together.

CODE:

```
#include <iostream>
using namespace std;

class Student {
private:
    string name;
    int marks;

    static int totalStudents;

public:
    Student(string n, int m) {
        name = n;
        marks = m;
        totalStudents++;
        cout << "Student " << name << " added. Total students: "
             << totalStudents << endl;
    }
    ~Student() {
        cout << "Student " << name << " removed from system" << endl;
        totalStudents--;
    }

    void display() {
        cout << "Name: " << name << ", Marks: " << marks << endl;
    }

    static int getTotalStudents() {
        return totalStudents;
    }
};

int Student::totalStudents = 0;

int main() {
    Student* students[10];
    int n;
```

Figure 1:CODE1

```
cout << "Enter number of students (max 10): ";
cin >> n;

if (n > 10) {
    cout << "Limit exceeded! Setting to 10.\n";
    n = 10;
}

for (int i = 0; i < n; i++) {
    string name;
    int marks;

    cout << "\nEnter name of student " << i + 1 << ": ";
    cin >> name;
    cout << "Enter marks: ";
    cin >> marks;

    students[i] = new Student(name, marks);
}

cout << "\n--- Displaying Student Info ---" << endl;
for (int i = 0; i < n; i++) {
    students[i]->display();
}

cout << "\n--- Deleting Students ---" << endl;
for (int i = 0; i < n; i++) {
    delete students[i];
    cout << "Total students left: "
        << Student::getTotalStudents() << endl;
}

return 0;
```

Figure 2:CODE 1

OUTPUT:

```
Enter number of students (max 10): 10
Enter name of student 1: ali
Enter marks: 67
Student ali added. Total students: 1

Enter name of student 2: ahmad
Enter marks: 56
Student ahmad added. Total students: 2

Enter name of student 3: awais
Enter marks: 78
Student awais added. Total students: 3

Enter name of student 4: jutt
Enter marks: 67
Student jutt added. Total students: 4

Enter name of student 5: qasim
Enter marks: 79
Student qasim added. Total students: 5

Enter name of student 6: roshan
Enter marks: 98
Student roshan added. Total students: 6

Enter name of student 7: dogger
Enter marks: 76
Student dogger added. Total students: 7

Enter name of student 8: mughees
Enter marks: 98
Student mughees added. Total students: 8

Enter name of student 9: hassan
Enter marks: 87
Student hassan added. Total students: 9

Enter name of student 10: waseem
Enter marks: 90
Student waseem added. Total students: 10

--- Displaying Student Info ---
Name: ali, Marks: 67
```

Figure 3:OUTPUT

```
Student jutt added. Total students: 4

Enter name of student 5: qasim
Enter marks: 79
Student qasim added. Total students: 5

Enter name of student 6: roshan
Enter marks: 98
Student roshan added. Total students: 6

Enter name of student 7: dogger
Enter marks: 76
Student dogger added. Total students: 7

Enter name of student 8: mughees
Enter marks: 98
Student mughees added. Total students: 8

Enter name of student 9: hassan
Enter marks: 87
Student hassan added. Total students: 9

Enter name of student 10: waseem
Enter marks: 90
Student waseem added. Total students: 10

--- Displaying Student Info ---
Name: ali, Marks: 67
Name: ahmad, Marks: 56
Name: awais, Marks: 78
Name: jutt, Marks: 67
Name: qasim, Marks: 79
Name: roshan, Marks: 98
Name: dogger, Marks: 76
Name: mughees, Marks: 98
Name: hassan, Marks: 87
Name: waseem, Marks: 90

--- Deleting Students ---
Student ali removed from system
Total students left: 9
Student ahmad removed from system
Total students left: 8
```

Figure 4:OUTPUT

```
Student awais removed from system  
Total students left: 7  
Student jutt removed from system  
Total students left: 6  
Student qasim removed from system  
Total students left: 5  
Student roshan removed from system  
Total students left: 4  
Student dogger removed from system  
Total students left: 3  
Student mughees removed from system  
Total students left: 2  
Student hassan removed from system  
Total students left: 1  
Student waseem removed from system  
Total students left: 0
```

Figure 5:OUTPUT